Machine-checked fragments of a base-10 prime pattern

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Abstract

This report records the results obtained on top of a pre-existing framework, and keeps the two apart. The framework (the base-10 encoding of primes, the end-digit key, the twin segmentation and the colour scheme) is Elizabeth Johnson's and was published earlier. The results here are new and joint. The main ones are an exact count, $3^{n+1} - 2^{n+1}$, of the *admissible* residue words of length n, together with a proof that every word produced by consecutive twin primes is one of them, so that the count is an upper bound for the realised words; a proof that only the prime 5 ever obstructs; the equilibrium obstruction; the exact range of the coprimality test; and a description of the base-10 colour wheel in coordinates, $\mathbb{Z}/10 \cong \mathbb{Z}/2 \times \mathbb{Z}/5$, on which every twin pair above 5 avoids the neutral hue. Some supporting facts (digit exclusivity for twin pairs, the uniqueness of $3, 5, 7$, the carry identity) are classical or elementary and are included because the formalisation needs them. The census, admissibility at 5 of every realised word, the equilibrium obstruction and the wheel identities at $W = 10$ are formalised in Lean 4 with Mathlib, and every axiom footprint is reported.

Authorship

Elizabeth Johnson (prior work)	**Jointly (this report)**
$p = 10s + d$ encoding; inner nodes	the census $A(n) = 3^{n+1} - 2^{n+1}$; realised $\Rightarrow$ admissible
end-digit key $g \equiv d_2 - d_1 \pmod{10}$	the carry identity (elementary)
mod-9 twin classes	the obstruction table (mod 2, 3, 5)
twin segmentation; counterflow framing	the seven word classes
the colour scheme (attribution under verification; see note)	digit exclusivity; junction uniqueness (classical; proofs included)
rung ladder $D = (g \bmod 10)/2 + 1$	the equilibrium obstruction; the range of the coprimality test
	the wheel in coordinates; twins on the wheel
	the Lean formalisations

Citation. For the framework alone, cite her manuscript *EJ's Alternate Base Prime Number Pattern and the Twin Prime Counterflow Model*, DOI 10.5281/zenodo.23378596. For anything in this report, cite both authors and that manuscript.

Posting. Posted by Jeromie Beasley with Elizabeth Johnson's knowledge; she may fork it or request changes (2026-11-04). The split above cites her framework to her alone and the new results to both.

Attribution note. A related document, *Advanced Color Theory in Relations to Primes* (Elizabeth Johnson and Shaun Neer, October 2026), lists a second author. The attribution of the colour scheme and its mixing rules, credited here to Elizabeth Johnson, is still being verified with the authors, and will be corrected in a later revision if needed.

1 Notation and the framework

Primes and digits. Every prime $p > 5$ is written $p = 10s + d$, with *end digit* $d \in \{1, 3, 7, 9\}$ and *inner node* s.

Twin starts and gaps. A *twin start* is a prime a with $a + 2$ also prime. We list them in increasing order:

$$a_1 = 3, \quad a_2 = 5, \quad a_3 = 11, \quad a_4 = 17, \quad a_5 = 29, \ \ldots$$

and write $d_k = a_{k+1} - a_k$ for the gaps (always with an index, to distinguish them from end digits), so that $d_1 = 2$ and $d_2 = d_3 = 6$. Every twin start $a \geq 5$ satisfies $a \equiv 5 \pmod 6$, because a and $a + 2$ are both coprime to 6. Hence

$$d_k \equiv 0 \pmod 6 \qquad (k \geq 2). \tag{1}$$

We call a_k *regular* if $a_k \geq 11$, that is, $k \geq 3$. The first two twin starts, 3 and 5, are the *seed*: they share the prime 5.

Segments. A *segment* runs from $bg = a_k$ (a twin start) to $ep = a_{k+1} + 2$ (the end of the *next* twin). The primes strictly between bg and ep are its *interior primes*. The *twin-to-twin gap* is $\gamma_k = a_{k+1} - (a_k + 2)$.

Theorem 1 (two channels)**.** *For integers $p < q$ with $p = 10s_p + d_p$, $q = 10s_q + d_q$ and $0 \leq d_p, d_q \leq 9$,*

$$q - p = 10\,\Delta s + \Delta d, \qquad \Delta s = s_q - s_p, \quad \Delta d = d_q - d_p.$$

Consequently

$$\Delta s = \left\lfloor \frac{q - p}{10} \right\rfloor + \mathbb{1}[\Delta d < 0].$$

Proof. The first identity is substitution. For the second: $-9 \leq \Delta d \leq 9$, so $q - p = 10\Delta s + \Delta d$ gives floor $= \Delta s$ when $\Delta d \geq 0$ and $\Delta s - 1$ when $\Delta d < 0$. □

This is a substitution identity. The gap splits into a bounded digit channel and an unbounded node channel. Among consecutive primes above 5 in the first $5 \cdot 10^6$, the carry $\mathbb{1}[\Delta d < 0]$ fires on 39.73% of gaps.

2 The residue language of the twin gaps

2.1 Words and admissibility

By (1) a regular gap is $\equiv 0 \pmod 6$, so its last digit $r = d \bmod 10$ lies in $\{0, 2, 4, 6, 8\}$ and determines $d \bmod 30$:

r	0	2	4	6	8
$d \bmod 30$	0	12	24	6	18

Definition 2. A *word* of length n is $w = (r_1, \ldots, r_n) \in \{0, 2, 4, 6, 8\}^n$. Let δ_i be the residue mod 30 assigned to r_i by the table, put $D_0 = 0$ and $D_i = \delta_1 + \cdots + \delta_i$, and define the *offset set*

$$\Delta(w) = \bigcup_{i=0}^{n} \{D_i,\ D_i + 2\} \subseteq \mathbb{Z}/30.$$

The word is *admissible* if, for each $q \in \{2, 3, 5\}$, the reduction of $\Delta(w)$ mod q is not all of $\mathbb{Z}/q$. The word *realised from* a_k is $(d_k \bmod 10, \ldots, d_{k+n-1} \bmod 10)$.

If w is realised from a_k, the $2n+2$ numbers $a_k + D'_i$ and $a_k + D'_i + 2$, where the D'_i are the true partial sums $d_k + \cdots + d_{k+i-1}$, are the twin primes a_{k+i} and $a_{k+i} + 2$. Their offsets from a_k reduce to $\Delta(w)$ modulo 2, 3 and 5.

Theorem 3 (realised words are admissible)**.** *Every word realised from a regular twin start is admissible.*

Proof. Let $q \in \{2, 3, 5\}$. If $\Delta(w)$ covered $\mathbb{Z}/q$, then some member $a_k + D'_i$ or $a_k + D'_i + 2$ of the tuple would be $\equiv 0 \pmod q$. That member is prime, so it equals $q \leq 5$. But every member is at least $a_k \geq 11$, which is a contradiction. □

Remark 4 (the seed is a genuine exception). The hypothesis $a_k \geq 11$ cannot be dropped. From $a_2 = 5$ the twin starts $5, 11, 17$ realise the word $(6, 6)$, with offsets $\{0, 2, 6, 8, 12, 14\} \equiv \{0, 1, 2, 3, 4\} \pmod 5$. This word is not admissible, and the proof fails because the tuple contains the prime 5 itself. If windows were allowed to start at 5, the counts of realised words below would read $5, 20, 66, 212, 666$ instead of $5, 19, 65, 211, 665$.

Theorem 5 (only 5 obstructs)**.** *For every word,* $\Delta(w) \bmod 2 = \{0\}$ *and* $\Delta(w) \bmod 3 = \{0, 2\}$*. Hence a word is admissible if and only if* $\Delta(w) \bmod 5 \neq \mathbb{Z}/5$.

Proof. Every $\delta_i \equiv 0 \pmod 6$, so every $D_i \equiv 0$ modulo 2 and 3. The set $\bigcup_i \{D_i, D_i + 2\}$ therefore reduces to $\{0\}$ mod 2 and to $\{0, 2\}$ mod 3, and neither covers its modulus. □

2.2 The census

Theorem 6 (census)**.** *The number of admissible words of length n is*

$$A(n) = 3^{n+1} - 2^{n+1}.$$

Proof. Step 1: walk positions. Write $r_i = 2\ell_i$ with $\ell_i \in \{0, 1, 2, 3, 4\}$; this ℓ_i is the rung index, one less than the rung of the framework's ladder. Since $\delta_i \equiv r_i \pmod{10}$, we have $\delta_i \equiv 2\ell_i \pmod 5$, so $D_i \equiv 2c_i \pmod 5$ with

$$c_0 = 0, \qquad c_i = \ell_1 + \cdots + \ell_i \pmod 5.$$

The map $(\ell_1, \ldots, \ell_n) \mapsto (c_1, \ldots, c_n)$ is a bijection $(\mathbb{Z}/5)^n \to (\mathbb{Z}/5)^n$, with inverse $\ell_i = c_i - c_{i-1}$.

Step 2: reduce to positions. Let $V = \{c_0, \ldots, c_n\}$ be the set of visited positions. Then $\Delta(w) \bmod 5 = 2 \cdot \big(V \cup (V + 1)\big)$, and multiplication by 2 is a bijection of $\mathbb{Z}/5$. By Theorem 5, w is admissible iff $V \cup (V + 1) \neq \mathbb{Z}/5$.

Step 3: the cover lemma. $V \cup (V + 1)$ misses a point m iff $m \notin V$ and $m - 1 \notin V$, that is, iff $V \subseteq \{m + 1, m + 2, m + 3\}$. So w is admissible iff V lies in a consecutive triple. Since $0 = c_0 \in V$, the triple contains 0, and there are exactly three such triples:

$$t_0 = \{0, 1, 2\}, \qquad t_1 = \{4, 0, 1\}, \qquad t_2 = \{3, 4, 0\}.$$

Step 4: inclusion–exclusion. By Steps 1–3, $A(n)$ is the number of $(c_1, \ldots, c_n) \in (\mathbb{Z}/5)^n$ with all c_i in one of t_0, t_1, t_2. Each box t_j^n has 3^n elements. The pairwise intersections are $\{0, 1\}^n$, $\{0, 4\}^n$ and $\{0\}^n$, of sizes 2^n, 2^n and 1, and the triple intersection is $\{0\}^n$. Hence

$$A(n) = 3 \cdot 3^n - \big(2^n + 2^n + 1\big) + 1 = 3^{n+1} - 2^{n+1}.$$ □

Remark 7 (the converse is not claimed). The converse of Theorem 3, that every admissible word is realised from some regular twin start, is a statement of prime-tuples type and is **not** proved here. It has been checked computationally that for $n \leq 5$ the words realised from regular twin starts within the first $5 \cdot 10^6$ primes are exactly the admissible ones. Nothing beyond that range is claimed.

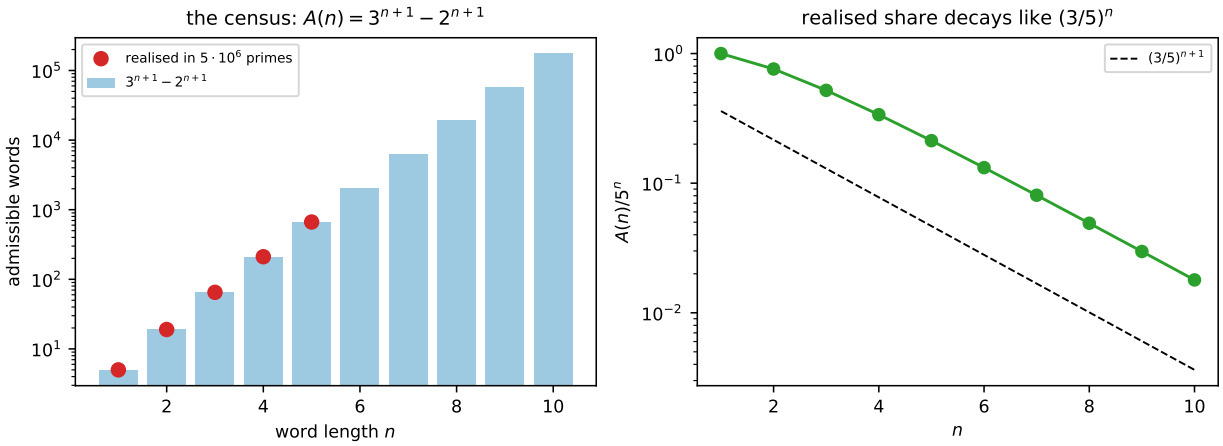


Figure 1: Left: the count $3^{n+1} - 2^{n+1}$ against the number of words realised from regular twin starts in the first $5 \cdot 10^6$ primes $(5, 19, 65, 211, 665)$. Right: the admissible share $A(n)/5^n = 3(3/5)^n - 2(2/5)^n$ decays like $(3/5)^n$, so almost all long residue words are impossible.

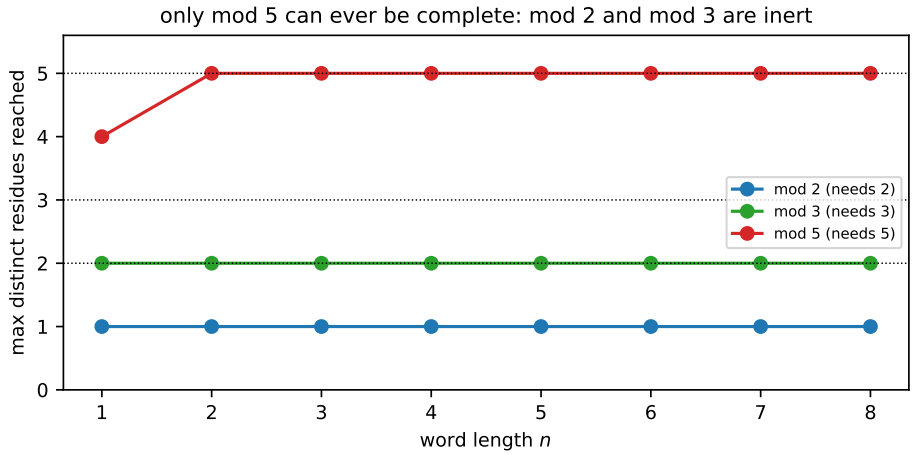


Figure 2: Maximum number of distinct residues reached, per modulus, over all words up to length 8. Only mod 5 can reach its modulus.

2.3 The seven word classes

Call $b_i = 2c_i \bmod 10 \in \{0, 2, 4, 6, 8\}$ the *beam* at step i. By Step 1 of the proof above, and because every D_i is even, $b_i \equiv D_i \pmod{10}$. In beam coordinates the three triples are $T_0 = \{0, 2, 4\}$, $T_1 = \{0, 2, 8\}$ and $T_2 = \{0, 6, 8\}$ (from t_0, t_1, t_2). We sort the admissible words by exactly which of these contain all their beams. The counts below follow from the same inclusion–exclusion and were also confirmed by enumeration for $n \le 6$.

class	beams	closed form	$n = 1$	2	3	4	5
T_0 only	$\subseteq \{0, 2, 4\}$	$3^n - 2^n$	1	5	19	65	211
T_1 only	$\subseteq \{0, 2, 8\}$	$3^n - 2^{n+1} + 1$	0	2	12	50	180
T_2 only	$\subseteq \{0, 6, 8\}$	$3^n - 2^n$	1	5	19	65	211
T_0T_1 only	$\subseteq \{0, 2\}$, not all 0	$2^n - 1$	1	3	7	15	31
T_1T_2 only	$\subseteq \{0, 8\}$, not all 0	$2^n - 1$	1	3	7	15	31
T_0T_2 only	—	0	0	0	0	0	0
$T_0T_1T_2$	all beams 0	1	1	1	1	1	1
total		$3^{n+1} - 2^{n+1}$	5	19	65	211	665

Three consequences follow.

- T_0 and T_2 have equal counts. This is because t_0 and t_2 each meet one neighbouring triple in two points and the other in one point, while t_1 meets both neighbours in two points.
- The beams 4 and 6 never occur in the same admissible word, since no triple containing 0 contains both 2 and 3. This is a statement about beams (positions). It is *not* a statement about letters: the word $(4, 6)$, with beams $4, 0$, is admissible, and the adjacent letters $4, 6$ or $6, 4$ occur 30 124 times in the data.
- Class "T_0T_2 only" is empty: $t_0 \cap t_2 = \{0\}$, and the all-zero word also lies in T_1.

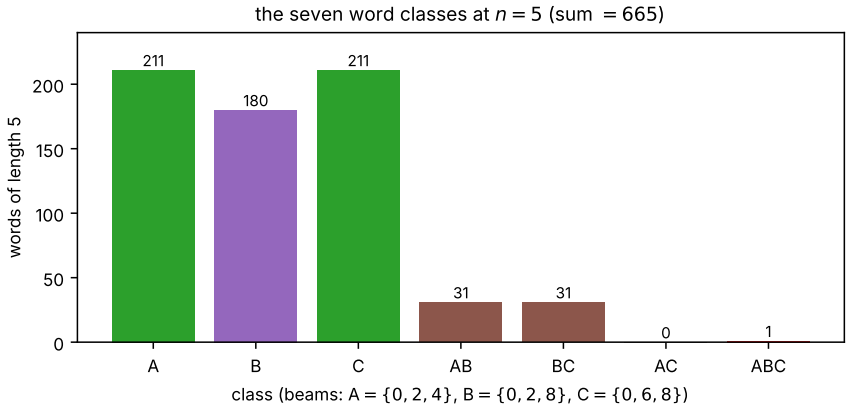


Figure 3: The seven classes at $n = 5$, summing to 665. The figure labels the classes T_0, T_1, T_2 as A, B, C.

3 Digits, junctions and equilibrium

Theorem 8 (digit exclusivity)**.** *Let $(a, a + 2)$ be a twin pair with $a > 5$. Then a ends in 1, 7 or 9, and $a + 2$ ends in 3, 9 or 1. In particular, a twin start ending in 3 is 3 itself, and a twin end ending in 7 is 7 itself.*

Proof. Since $a > 5$ is prime, a is odd and $5 \nmid a$. If $a \equiv 3 \pmod 5$ then $5 \mid a + 2$ with $a + 2 > 5$, which is impossible. So $a \bmod 5 \in \{1, 2, 4\}$, and since a is odd, $a \bmod 10 \in \{1, 7, 9\}$. Then $a + 2 \bmod 10 \in \{3, 9, 1\}$. For the last sentence: 3 and 7 are not among the allowed digits for $a > 5$, and the cases $a \leq 5$ are $(3, 5)$ and $(5, 7)$. □

Remark 9. Primes ending in 3 above 3, and primes ending in 7 above 7, both exist ($13, 23, 43$ and $17, 37, 47$). The theorem is about the two positions in a *twin pair*, not about all primes. The statement is classical; it is included because the formalisation uses it.

Theorem 10 (junction uniqueness)**.** *Among $p - 2, p, p + 2$ exactly one is divisible by 3. Consequently, if all three are prime then $p = 5$, and the twin-to-twin gap γ_k vanishes only for $k = 1$.*

Proof. The three numbers are pairwise distinct mod 3 because $2 \not\equiv 0 \pmod 3$, so exactly one is $\equiv 0$. If all three are prime, that one equals 3, so $p \in \{1, 3, 5\}$; only $p = 5$ makes all three prime. Finally, $\gamma_k = 0$ means $a_k + 2 = a_{k+1}$, so $a_k, a_k + 2, a_k + 4$ are all prime, and then $a_k = 3$. □

Theorem 11 (equilibrium obstruction)**.** *Say equilibrium holds at k if $d_k = d_{k+1}$, i.e. a_k, a_{k+1}, a_{k+2} are in arithmetic progression. Let a_k be regular, and let $u = a_k \bmod 10$ and $v = a_{k+1} \bmod 10$ (both in $\{1, 7, 9\}$ by Theorem 8). If equilibrium holds at k, then*

$$a_{k+2} \equiv 2v - u \pmod{10} \quad \text{and} \quad 2v - u \bmod 10 \in \{1, 7, 9\}.$$

Of the nine cells $u \to v$, exactly four violate this: $1 \to 7$, $7 \to 1$, $9 \to 1$ and $9 \to 7$. In those cells equilibrium is impossible.

Proof. $a_{k+2} = a_{k+1} + d_{k+1} = a_{k+1} + d_k = 2a_{k+1} - a_k$, so $a_{k+2} \equiv 2v - u \pmod{10}$. Since $a_{k+2} > a_k \geq 11$ is a twin start, its last digit lies in $\{1, 7, 9\}$ by Theorem 8. The table lists the nine cells. □

cell $u \to v$	$r = v - u \bmod 10$	$2v - u \bmod 10$	equilibrium
$1 \to 1$, $7 \to 7$, $9 \to 9$	0	1, 7, 9	possible
$7 \to 9$	2	1	possible
$9 \to 1$	2	3	impossible
$7 \to 1$	4	5	impossible
$1 \to 7$	6	3	impossible
$1 \to 9$	8	7	possible
$9 \to 7$	8	5	impossible

In terms of the gap residue $r = d_k \bmod 10$: the residues 4 and 6 never equilibrate; the residues 2 and 8 equilibrate only from the single cells $7 \to 9$ and $1 \to 9$ respectively; and the residue 0 can equilibrate from all three of its cells. Over regular twin starts in the first $5 \cdot 10^6$ primes, the four impossible cells show exactly zero equilibria, and the five possible cells show equilibrium rates between 1.66% and 2.48%. The spread among the possible cells is studied empirically in `03-docs/THEOREMS.md`; it is not a theorem of this report.

Notation note. The framework documents label a cell by *bg* → *ep*, the end digit of a_k and of $a_{k+1} + 2$, i.e. $u \to v + 2$. In that notation the impossible cells are $1 \to 9$, $7 \to 3$, $9 \to 3$ and $9 \to 9$.

4 The range of the coprimality test

Theorem 12. *Let P be a product of distinct primes, and let p be the smallest prime not dividing P. Let $n > 1$ be an integer.*

(a) If $\gcd(n, P) > 1$ and n is not itself a prime factor of P, then n is composite;

(b) if $\gcd(n, P) = 1$ and $n < p^2$, then n is prime;

and $n = p^2$ is the smallest $n > 1$ with $\gcd(n, P) = 1$ that is composite.

Proof. (a) A prime factor q of P divides n, and $n \neq q$. (b) Every prime below p divides P, so every prime factor of n is at least p. A composite n would therefore satisfy $n \geq p^2$. Finally, p^2 is coprime to P and composite, and by (b) nothing smaller is. □

The end digit tests coprimality to 10: $n \bmod 10 \in \{1, 3, 7, 9\}$ iff $\gcd(n, 10) = 1$. On its own ($P = 10$, $p = 3$) it confirms primality only below 9. Adding the test $3 \nmid n$ gives the 30-wheel ($P = 30$, $p = 7$), which confirms primality exactly up to $n < 49$; $49 = 7^2$ is the first failure. Every such finite test therefore refutes reliably but confirms only below p^2. Deciding larger n requires enlarging P as n grows, which is trial division.

5 The colour wheel

The colour scheme of the framework labels the ten digits: primaries $1, 3, 7, 9$, secondaries $2, 4, 6, 8$, grey 5, base 0. This section shows how much of that labelling is intrinsic to the ring $\mathbb{Z}/10$ and how much is convention. The orbit partition, the two layers, the hue coordinate and complement as negation are forced. The direction of the hue circle, and which orbit is called primary, are choices. It then places the twin primes on the resulting wheel. The words *layer*, *hue* and *grey* below are names for coordinates. No colorimetric or physical claim is made (Remark 18).

5.1 Orbits

Theorem 13 (orbits)**.** *Let $W \geq 2$. The orbits of the unit group $(\mathbb{Z}/W)^\times$ acting on $\mathbb{Z}/W$ by multiplication are the classes $\{x : \gcd(x, W) = g\}$, one for each divisor $g \mid W$. The class with*

$W/g = m$ has $\varphi(m)$ elements. At $W = 10$ the four orbits are

$$\{0\}, \qquad \{5\}, \qquad \{2,4,6,8\}, \qquad \{1,3,7,9\}.$$

Proof. Multiplying by a unit does not change $\gcd(x, W)$. Conversely, suppose $\gcd(x, W) = \gcd(y, W) = g$, and put $m = W/g$, $x = gx'$, $y = gy'$ with x', y' units mod m. Choose t with $tx' \equiv y' \pmod m$. Let W' be the largest divisor of W coprime to m. By the Chinese remainder theorem there is u with $u \equiv t \pmod m$ and $u \equiv 1 \pmod{W'}$. Every prime factor of W divides m or W', so u is a unit mod W. Then $ux - y = g(ux' - y')$ and $m \mid ux' - y'$, so $W \mid ux - y$. For the size: the orbit of x consists of the residues $g\,x''$ with x'' a unit mod m, and $g\,x''$ determines x'' mod m. So the orbit has $\varphi(m)$ elements. □

Proposition 14. *The orbit sizes are $1, 1, m, m$ for some m if and only if $W = 2p$ with p an odd prime, in which case $m = p - 1$. This gives the family $W = 6, 10, 14, 22, 26, 34, 38, \ldots$*

Proof. Orbits correspond to divisors, so W has exactly four divisors: $W = q^3$ or $W = qr$ with primes $q < r$. For $W = q^3$ the sizes are $1, q-1, q(q-1), q^2(q-1)$; two of them equal 1 only if $q = 2$, which gives $1, 1, 2, 4$, not of the required shape. For $W = qr$ the sizes are $1, q-1, r-1, (q-1)(r-1)$; two ones force $q = 2$, and then the sizes are $1, 1, r-1, r-1$. Conversely, $W = 2p$ gives these sizes. □

5.2 The wheel in coordinates

Theorem 15 (coordinates)**.** *Let $W = 2p$ with p an odd prime. Write $\lambda(x) = x \bmod 2$ (the* layer*) and $h(x) = x \bmod p$ (the* hue*). Then $x \mapsto (\lambda(x), h(x))$ is a ring isomorphism $\mathbb{Z}/2p \to \mathbb{Z}/2 \times \mathbb{Z}/p$, and the following hold.*

(a) The four orbits are $\{(0,0)\}$, $\{(1,0)\}$ (the grey *point p), $\{(0,h) : h \neq 0\}$ and $\{(1,h) : h \neq 0\}$.*

(b) A unit u acts by $(\lambda, h) \mapsto (\lambda, \bar{u}\,h)$ with $\bar{u} = u \bmod p$, and $u \mapsto \bar{u}$ is an isomorphism $(\mathbb{Z}/2p)^\times \to (\mathbb{Z}/p)^\times$, a cyclic group of order $p-1$. Fix a generator γ of $(\mathbb{Z}/p)^\times$ and define the angle *of a non-grey hue by $h = \gamma^\theta$, $\theta \in \mathbb{Z}/(p-1)$. Then every unit acts on the non-neutral points of each layer as a rotation $\theta \mapsto \theta + j$, simply transitively. The non-neutral points of each layer therefore form a circle of $p-1$ hues.*

(c) Complementation $x \mapsto -x$ is the half turn $\theta \mapsto \theta + \frac{p-1}{2}$, and it preserves layers.

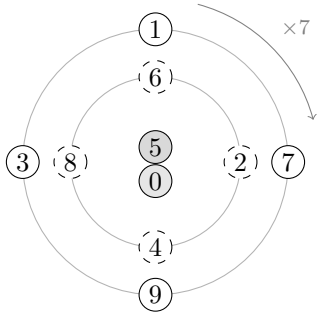
(d) Adding the grey point, $x \mapsto x + p$, swaps the two layers and keeps the hue.

Proof. The map is the Chinese remainder isomorphism. (a) This is Theorem 13 in coordinates. For instance, $\gcd(x, 2p) = 1$ iff $\lambda = 1$ and $h \neq 0$. (b) A unit is odd, so it fixes λ. The reduction map is a bijection because a unit is determined by its residue mod p, its layer being 1. Cyclicity holds because p is prime. (c) $-1 = \gamma^{(p-1)/2}$ in $(\mathbb{Z}/p)^\times$. (d) p corresponds to $(1, 0)$. □

Remark 16 (what depends on the generator). The angle θ depends on the choice of γ. Replacing γ by γ^j, with j a unit mod $p-1$, multiplies every angle by j^{-1}. Up to rotation and reflection this gives $\varphi(p-1)/2$ different cyclic orders of the hues. That is one order for $p = 5$ and 7, where only the orientation is a choice, and two for $p = 11$ and 13, which we checked by enumeration. The partition into orbits, the layers, the hue coordinate and the half turn $x \mapsto -x$ do not depend on any choice.

At $W = 10$ take $\gamma = 2$; it is realised by the unit 7, since $7 \equiv 2 \pmod 5$. Multiplication by 7 is the quarter turn:

angle θ	0	1	2	3
hue $h = 2^\theta \bmod 5$	1	2	4	3
odd layer (primaries)	1	7	9	3
even layer (secondaries)	6	2	4	8



So at $W = 10$ the scheme of the framework is recovered from the ring structure, up to the orientation of the circle and the choice of which orbit to call primary. There is a neutral axis $\{0, 5\}$. Each layer has four non-neutral points arranged as a circle of hues, with the framework's primaries on the unit layer. Each primary x has a secondary partner $x + 5$ of the same hue: $1-6$, $7-2$, $9-4$, $3-8$. The complementary pairs $\{1, 9\}, \{7, 3\}, \{6, 4\}, \{2, 8\}$ are exactly the opposite points of each circle, and complementary digits add to the base: $x + (-x) = 0$.

5.3 Twin primes on the wheel

Proposition 17 (twins avoid grey)**.** *Let $(a, a+2)$ be a twin pair with $a > 5$. Both members lie on the unit layer and have non-grey hue. More precisely, $h(a) \in \{1, 2, 4\}$ (angles $0, 1, 2$; digits $1, 7, 9$) and $h(a + 2) = h(a) + 2 \in \{3, 4, 1\}$ (angles $3, 2, 0$; digits $3, 9, 1$). The same argument at any odd prime $q < a$ gives $a \bmod q \notin \{0,\, q - 2\}$.*

Proof. Both members are primes greater than 5, hence odd and coprime to 5. $h(a) \neq 0$, and $h(a + 2) = h(a) + 2 \neq 0$ forces $h(a) \neq 3$. The general statement is the same argument mod q. □

This is Theorem 8 in wheel coordinates, and it connects the two halves of the report. The twin step $x \mapsto x + 2$ *translates* the hue additively. Theorem 15 shows that the wheel symmetries are *multiplications*, which rotate hue. Twin structure therefore lives in the additive structure of the ring and the wheel symmetry in the multiplicative one. The census has the same reading. Every member of a realised tuple has a non-grey hue. Admissibility at 5 says that the offset pattern $\bar{\Delta}(w) = \Delta(w) \bmod 5$ can be translated off the grey hue: there is an $h_0 \in \mathbb{Z}/5$ with $0 \notin h_0 + \bar{\Delta}(w)$. This is immediate: take $h_0 = -m$ for any $m \notin \bar{\Delta}(w)$. Theorem 6 thus counts the *words* whose hue pattern can be translated off grey. Different words can share a pattern.

Remark 18 (what the wheel does not do). The orbits classify a *label system* on $\mathbb{Z}/W$. Nothing here is a claim about perceived or physical colour, and the words layer, hue and grey are names for coordinates. On prime gaps: a unit u permutes the residues coprime to W, so it fixes the set of wheel survivors. We checked by enumeration at $W = 10, 30, 210, 2310$ that only $u = 1$ preserves the cyclic order of the survivors and only $u = -1$ reverses it. For instance, the 30-wheel gap cycle $6, 4, 2, 4, 2, 4, 6, 2$ is a palindrome up to rotation. The unit multiplications therefore contribute exactly the reflection symmetry of the wheel's gap pattern. They act on residues only, not on the primes themselves.

5.4 Colour at scale

"Scale" has two meanings here, and the arithmetic treats them differently. *Resolution* means reading more digits: colouring n by $n \bmod 10^k$ instead of $n \bmod 10$. *Magnitude* means looking at larger numbers. The first is exact. The second is statistical.

Resolution: the tower of wheels. Reduction mod 10^k is a ring map $\pi : \mathbb{Z}/10^{k+1} \to \mathbb{Z}/10^k$, the *projection* from one scale to the next coarser one. By Theorem 13 the colour classes of the 10^k-wheel are the gcd classes $g = 2^i 5^j$, $0 \leq i, j \leq k$, so there are $(k + 1)^2$ of them.

Proposition 19 (classes refine)**.** *Projection sends each class of $\mathbb{Z}/10^{k+1}$ into a single class of $\mathbb{Z}/10^k$, namely the one with* gcd *equal to* $\gcd(g, 10^k)$*. Over the four classes of the* 10*-wheel, the 10^k-wheel has*

$$\underbrace{1}_{\textit{primaries}} + \underbrace{k}_{\textit{secondaries}} + \underbrace{k}_{\textit{grey}} + \underbrace{k^2}_{\textit{base}} = (k+1)^2$$

classes, the shades *of each colour. In particular the primaries never split: n is a unit mod 10^k iff it is a unit mod* 10*. Every prime above* 5 *lies in this single class at every scale.*

Proof. $\gcd(x \bmod 10^k, 10^k) = \gcd(x, 10^k) = \gcd(\gcd(x, 10^{k+1}), 10^k)$. Iterating down to scale 10: a class $2^i 5^j$ lies over the primaries iff $i = j = 0$; over the secondaries iff $i \geq 1$, $j = 0$ (k choices); over grey iff $i = 0$, $j \geq 1$ (k choices); and over the base iff $i, j \geq 1$ (k^2 choices). □

At scale 100, for example, the secondaries split into two shades (divisible by 2 only, and by 4). Grey splits into two (divisible by 5 only, and by 25, i.e. 25 and 75). The base splits into four. The hue of the unit class is the coordinate $n \bmod 5^k$. Its circle is the cyclic group $(\mathbb{Z}/5^k)^\times$, which has $N_k = 4 \cdot 5^{k-1}$ elements: 4 hues at scale 10, 20 at scale 100, 100 at scale 1000.

Definition 20. A *hue circle* at scale 10^k is the cyclic order $x, gx, g^2x, \ldots$ on $(\mathbb{Z}/5^k)^\times$ given by a generator g; it does not depend on the starting point x. The circle of h at scale 10^{k+1} *refines* the circle of g at scale 10^k if $h \equiv g \pmod{5^k}$, equivalently if projection sends each step $y \mapsto hy$ to the step $\pi(y) \mapsto g\,\pi(y)$.

Proposition 21 (zooming keeps the coarse order)**.** *Let g generate $(\mathbb{Z}/5^k)^\times$. Then $g \bmod 5$ generates $(\mathbb{Z}/5)^\times$, so $g \equiv 2$ or $3 \pmod 5$. Projected to scale 10, the circle of g is the coarse circle of 2, traversed 5^{k-1} times: forwards if $g \equiv 2$, backwards if $g \equiv 3$. In particular a circle that refines the coarse circle of 2, in the sense of Definition 20 iterated, runs forwards.*

Proof. Reduction $(\mathbb{Z}/5^k)^\times \to (\mathbb{Z}/5)^\times$ is a surjective homomorphism (every unit mod 5 lifts), so it sends a generator to a generator. The generators mod 5 are 2 and $3 = 2^{-1}$. If $g \equiv 2^{\pm 1} \pmod 5$, then from any start x the coarse hue of the θ-th step is $\bar{x}\, 2^{\pm\theta}$. That expression advances the coarse angle by ± 1 at every step, and the $N_k = 4 \cdot 5^{k-1}$ steps of the fine circle make 5^{k-1} coarse turns. □

Proposition 22 (how much freedom each zoom adds)**.**

(a) At scale 10^k there are $\varphi(N_k)$ hue circles, or $\varphi(N_k)/2$ up to reflection: 1, 4, 20 for $k = 1, 2, 3$.

(b) A given circle at scale 10 has exactly 4 refinements at scale 100. Every circle at scale 10^k with $k \geq 2$ has exactly 5 refinements at scale 10^{k+1}.

(c) The circles of 2 refine one another at every scale: 2 generates $(\mathbb{Z}/5^k)^\times$ for every k.

Proof. (a) The circle of g determines g as the successor of 1, and reading the circle backwards gives the circle of $g^{-1} \neq g$ (as $N_k > 2$). So there is one circle per generator. (b) The refinements of the circle of g are the generators mod 5^{k+1} among the five lifts $g + j5^k$. A unit mod p^{k+1}, with p an odd prime and $k \geq 2$, is a generator iff its reduction mod p^2 is [1]. All five lifts have the same reduction mod 25 as g, so all five are generators. For $k = 1$: the solutions of $x^4 \equiv 1 \pmod{25}$ form a subgroup of order 4 with exactly one element over each unit mod 5 ($1, 7, 18, 24$ over $1, 2, 3, 4$). A lift of a generator mod 5 has order 4 or 20, and it has order 4 exactly when it is that element. So exactly 4 of the 5 lifts generate. For $g = 2$ the exception is 7. (c) 2 has order 20 mod 25, and a primitive root mod p^2 is a primitive root mod every p^k [1]. □

At scale 100 the four refinements of the coarse circle of 2 start, after 1, with $2, 4, 8, 16$, or $12, 19, 3, 11$, or $17, 14, 13, 21$, or $22, 9, 23, 6$. Each reads the coarse hues $1, 2, 4, 3$ five times round. So zooming in preserves the coarse cyclic order exactly once an orientation is fixed, and it still leaves real freedom in the fine order: a finite set of refinements at each step (4 at the first zoom, then 5 at every later zoom). The circle of 2 is a canonical choice that is coherent at every scale.

Two cautions. First, the unit class carries more than its hue: $(\mathbb{Z}/100)^\times \cong C_2 \times C_{20}$, where the extra C_2 is $n \bmod 4$. That layer coordinate $n \bmod 2^k$ refines into $(\mathbb{Z}/2^k)^\times$, which is *not* cyclic for $k \geq 3$ ($(\mathbb{Z}/8)^\times \cong C_2 \times C_2$). So from scale 1000 on, the layer side does not form a single circle. Second, hue circles are defined here for the unit class only. How to give hue to the other shades is open.

Magnitude: the colour proportions drift. By the prime number theorem for arithmetic progressions, each of the four primaries ends exactly 1/4 of the primes asymptotically. At any finite size the proportions are not equal, and the following are *measurements* over the first $5 \cdot 10^6$ primes above 5, not theorems.

- *A persistent lead.* The primes ending in 3 or 7 (the non-squares mod 10) were never strictly behind those ending in 1 or 9 (the squares), checked at every prime. They were tied only at $p = 11, 31, 41, 3581$. This is Chebyshev's bias. Under the generalised Riemann hypothesis and a linear-independence hypothesis, Rubinstein and Sarnak [3] showed that such a lead

holds on a set of logarithmic density strictly between 0 and 1. So it is expected to fail eventually, on a small but positive proportion of the logarithmic scale: it is persistent, not permanent.

- *An order that churns.* Ranking the four digits by count at 60 logarithmically spaced checkpoints gives 49 checkpoints with no tie, and the ranking changes 36 times between consecutive ones. Seven different rankings occur. Checked at every prime, 3 is strictly last in four stretches, starting at $p = 3571$, 55 619, 55 639 and 56 311.
- *A fading repulsion.* The share of consecutive primes with the same last digit was 11.84% below 10^4, 15.35% below 10^6 and 17.12% over all $5 \cdot 10^6$ primes, against 25% for independent digits. Lemke Oliver and Soundararajan [2] found this bias and gave a conjectural explanation through the Hardy–Littlewood conjectures, in which it decays very slowly, roughly like $1/\log x$. It lives exactly in the sixteen end-digit cells of the framework, which is why no cell count in this work is treated as a constant.

These figures are computed by `04-verification/scale_check.py`.

Summary. Under zooming, the coarse cyclic order is preserved exactly, once an orientation is chosen, and each zoom adds a finite, explicitly counted freedom in the fine order (Propositions 21–22). Under growth there is a persistent but not permanent bias, with the finer ranking churning beneath it. The theory of that bias is conditional (Rubinstein–Sarnak) or conjectural (Lemke Oliver–Soundararajan). Both kinds of scale show a coarse structure that persists while the finer order moves, but only the first is a theorem.

5.5 What a theory built on this seed has, and what it still needs

Established here. The wheel $\mathbb{Z}/2 \times \mathbb{Z}/p$ has a neutral axis and two layers, each with a circle of $p-1$ hues on which the units act by rotation, and complement is the half turn. The partition and coordinates are forced by the ring, and the cyclic order of the hues is forced up to the choices in Remark 16. The primes above p occupy the unit layer. Twin pairs above q avoid hue 0 and hue -2 at every odd prime q. The census counts the words whose hue pattern can be translated off grey. Across scales, the wheels form a tower in which classes refine and the primaries never split. Each zoom preserves the oriented coarse hue order and adds a finite, counted freedom in the fine order (Section 5.4).

Open, and stated as open.

(i) *A mixing law.* The ring supplies two candidate operations: addition, which adds hues and layers, and multiplication, which rotates hues. Which of them, if either, should model "mixing" is a modelling decision, not a theorem. Under addition, complements mix to the base, as noted above.

(ii) *Finer wheels.* Proposition 14 gives one wheel for each odd prime. Every admissible word can be lifted to gap sequences that are admissible at every prime: by the Chinese remainder theorem, choose the gaps $\equiv 0$ modulo all primes ≥ 7 up to any bound. What is open is the converse of Theorem 3, which needs two things. One is a prime k-tuples statement, that such tuples actually occur. The other is consecutiveness: no further twin pair may fall between the members. The Hardy–Littlewood conjecture alone does not give the second.

(iii) *Rates.* Whether the measured equilibrium rates (Section 3) can be expressed through hue statistics is untested.

(iv) *Choosing the order at every scale.* Proposition 22 says what the choices are. Whether the framework wants the canonical circle of 2, or some other coherent family, is a modelling decision. So is how to treat the non-cyclic layer side from scale 1000 on.

(v) *Shades.* Proposition 19 gives the shades of each colour at every scale. Whether shades should carry their own order (for instance by the exponents i, j) is open.

(vi) *Delays beyond 5.* The free-hue weights ω_q at every prime, and the correction for consecutiveness (Section 6), are the natural next quantitative target.

6 Delays on the wheel

A *delay* is the waiting time until something recurs. This section treats two kinds. The first is the delay, counted in primes, until an end digit comes back. The second is the delay, measured in integers, between twin pairs. The rule for the second is exact; how well it describes the primes is measured, and every measurement below is computed by `04-verification/delay_check.py` on the first $5 \cdot 10^6$ primes above 5.

6.1 Return delays of a colour

For a sequence in which a colour has frequency f, the mean delay between its occurrences is $1/f$ up to boundary terms. This is because the delays sum to the distance from the first occurrence to the last. Each primary has frequency $1/4$ asymptotically (Section 5.4), so its mean return delay is 4. The data give 4.001, 3.999, 4.000, 4.000 for the digits $1, 3, 7, 9$. The *rate fixes the mean, but not the shape*:

delay (primes)	1	2	3	4	5
digit 3	16.59%	21.42%	17.19%	12.63%	9.16%
digit 1	17.65%	20.00%	17.22%	12.76%	9.26%
independent digits	25.00%	18.75%	14.06%	10.55%	7.91%

An immediate return is strongly suppressed, and delays 2 to 5 are correspondingly enhanced. This is the bias of Lemke Oliver and Soundararajan [2] seen as a waiting time. Its range is short. The mutual information between a digit and the digit k primes later is 36.5, 2.0 and 0.11 millibits for $k = 1, 2, 3$ (out of an entropy of 2 bits per digit), and below 0.15 millibits for $k \geq 3$. So the memory of the colour sequence lasts about two primes.

6.2 Twin delays are weighted by free hues

Proposition 23 (free hues)**.** *Let h be the delay between two twin pairs $(a, a+2)$ and $(a+h, a+h+2)$. The number of hues $a \bmod 5$ for which all four members avoid the grey hue 0 is*

$$\omega_5(h) = 5 - \big|\{0, 2, h, h+2\} \bmod 5\big| \;=\; 3,\ 1,\ 2,\ 2,\ 1 \qquad \text{for } h \equiv 0, 1, 2, 3, 4 \pmod 5.$$

More generally, at any prime $q \geq 3$ the count is $\omega_q(h) = q - |\{0, 2, h, h+2\} \bmod q|$.

Proof. A hue a fails exactly when $a \equiv -u$ for some u in the offset pattern $\{0, 2, h, h+2\}$, and distinct offsets give distinct failing hues. List the five patterns mod 5 to get the values. □

$\omega_5(h)$ is exactly the number of translations of the offset pattern off grey (Section 5.4, and the census reading after Proposition 17). It is also the local factor at 5 of the Hardy–Littlewood conjecture for the tuple $\{0, 2, h, h+2\}$. That conjecture predicts that the number of twin pairs at distance h (not necessarily consecutive) grows in proportion to the singular series $\mathfrak{S}(h) = \prod_q \big(1 - \nu_q(h)/q\big)(1 - 1/q)^{-4}$, where $\nu_q(h) = q - \omega_q(h)$ is the number of residues the pattern occupies. The factor at $q = 5$ is proportional to $\omega_5(h)$. For delays between twin starts above 5, $h \equiv 0 \pmod 6$, so the factors at 2 and 3 do not depend on h. Averaging the factors at $q \geq 7$ over h in each class mod 5, it predicts shares proportional to $3 : 1 : 2 : 2 : 1$, i.e. 33.3%, 11.1%, 22.2%, 22.2%, 11.1%. This is a prediction of a conjecture, not a theorem. Measured:

delay $h \bmod 5$	0	1	2	3	4
free hues $\omega_5(h)$	3	1	2	2	1
predicted share	33.3%	11.1%	22.2%	22.2%	11.1%
all pairs of twin starts, $h \leq 3000$	33.1%	11.1%	22.3%	22.2%	11.2%
consecutive twin pairs	30.7%	12.1%	24.1%	22.1%	11.1%

Over $5.27 \cdot 10^6$ pairs of twin starts at distance at most 3000, the free-hue weights fit to within 0.2 percentage points. The consecutive delays deviate by up to 2.6 points. This is expected: requiring that *no* twin pair falls in between is the consecutiveness condition that the plain conjecture does not cover (Section 5.4, open item (ii)). In colour language, a delay happens more often the more hue room it leaves for the pair to avoid grey.

6.3 Two checks that came out null

The *hue walk*, stepping round the circle of 2 from each prime to the next, has net drift +1654 quarter turns over $5 \cdot 10^6$ steps. That is $z = 0.99$ against a walk without drift, so there is no detectable rotation. *Equilibria* ($d_k = d_{k+1}$, Theorem 11) follow one another within five twin pairs 168 times, against 192.8 for independent equilibria at the same rate ($z = -1.79$). That is a hint of mutual avoidance, and it is not significant at this size. Both are recorded so that they are not rediscovered as findings.

7 Formalisation

The following are machine-checked in Lean 4 (v4.33.0) against Mathlib (commit `db584cd`, checked out from tag `v4.33.0`). The three files share one small list of Mathlib imports rather than importing all of Mathlib. All three were compiled for this revision with no errors, warnings or `sorry`. The printed axiom footprints are recorded verbatim in `05-handoff/manifest.json`.

file	content	axioms
`Census.lean`	`census_words`: Theorem 6 on words, for all n, including the bridge from words to walk positions (`cover_iff`, `admissible_iff`); `realised_admissible`: Theorem 3 at $q = 5$ for $a > 5$, with the seed counterexample of Remark 4 checked; `census`: the inclusion–exclusion count	`propext`, `Classical.choice`, `Quot.sound`
`EJPrime.lean`	twin-start digits and digit exclusivity (Theorem 8); junction (Theorem 10, the "all prime $\Rightarrow$ $p = 5$" part); `equilibrium_obstruction` (Theorem 11, stated on primes; it assumes only that $2b - a$ and $2b - a + 2$ are prime, which is more general), plus the nine cell identities	`propext`, `Quot.sound`; cell identities `propext` only
`Wheel.lean`	at $W = 10$, checked by `decide`: layer and hue determine a digit (injectivity of the coordinate map); orbits are the gcd classes (Theorem 13); units fix the layer; the ten values of $\times 7$ (the quarter turn); $9x = -x$; the layer swap. Also, on primes: twin starts above 5 have hue in $\{1, 2, 4\}$ (Proposition 17 at 5)	coordinates: none; orbits, layer: `propext`, `Classical.choice`, `Quot.sound`; $\times 7$, $9x = -x$, swap: `propext`; twin lemmas: `propext`, `Quot.sound`
	scale (by `decide`, with explicit witnesses for the nine classes): coarse class is a function of fine class at 100; the nine classes of $\mathbb{Z}/100$; 2 generates mod 25; the fine circle of 2 projects to the coarse one; the eight generators mod 25 and their reductions; the five lifts of 2 to mod 125 all generate	at most `propext`, `Quot.sound`
	delays: `free_hues` ($\omega_5 = 3, 1, 2, 2, 1$) and `free_hues_eq` ($\omega_5 = 5-$ pattern size), Proposition 23 at $q = 5$	`propext`, `Classical.choice`, `Quot.sound`

Not formalised: Theorem 1; Theorem 5. Lean defines admissibility at 5 only, so `census_words` is Theorem 6 through Theorem 5, and that reduction is on paper. Also not formalised are Theorem 12; the general-W statements of Theorems 13 and 15 and Proposition 14, of which only $W = 10$ facts are checked; the ring-isomorphism and half-turn statements beyond those $W = 10$ identities; Remark 16; the general-k statements of Propositions 19–22, of which only the scale-100 and scale-125 facts above are checked (in particular the circle counts $\varphi(N_k)/2$ are on paper only); Proposition 23 for $q \neq 5$; and every measured rate.
The hypotheses of `realised_admissible` are: $a > 5$; gaps $d_1, \ldots, d_n$ that are all even; and every $a + D'_i$ and $a + D'_i + 2$ prime, where $D'_i = d_1 + \cdots + d_i$ are the true partial sums (Lean's `Dn`). It requires neither consecutive twin starts nor $d_i \equiv 0 \pmod 6$, so it is more general than Theorem 3 at $q = 5$.

8 Scope

- The converse of Theorem 3 is not claimed. It is observed for $n \leq 5$.
- No finite test decides primality in general; see Theorem 12.
- The wheel symmetries return no information about prime gaps (Remark 18).
- No physical or colorimetric claim is made anywhere in this report.
- The seed twin starts 3 and 5 are exceptions wherever stated (Remark 4, Theorems 8–11), and every formal statement carries the corresponding hypothesis.

References

[1] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Springer, 1990, Ch. 4 (primitive roots modulo prime powers).

[2] R. J. Lemke Oliver and K. Soundararajan, Unexpected biases in the distribution of consecutive primes, *Proc. Natl. Acad. Sci. USA* 113 (2016), E4446–E4454.

[3] M. Rubinstein and P. Sarnak, Chebyshev's bias, *Experimental Mathematics* 3 (1994), 173–197.

Reproduction

All numbers were computed from a sieve of the first 5 000 000 primes above 5 (up to 86 028 223) in Python with NumPy (`04-verification/verify_all.py`: 24/24 checks pass; the scale measurements in Section 5.4 come from `04-verification/scale_check.py`, and those in Section 6 from `04-verification/delay_check.py`). The figures come from `04-verification/make_figs.py`. For the Lean files, run `lake env lean <file>` inside a Lake project that depends on Mathlib v4.33.0.

Statements were developed and checked with AI assistants under human direction, including independent AI referee passes on revisions 4, 5 and 6. As with any tool-assisted work, responsibility for correctness rests with the named authors.